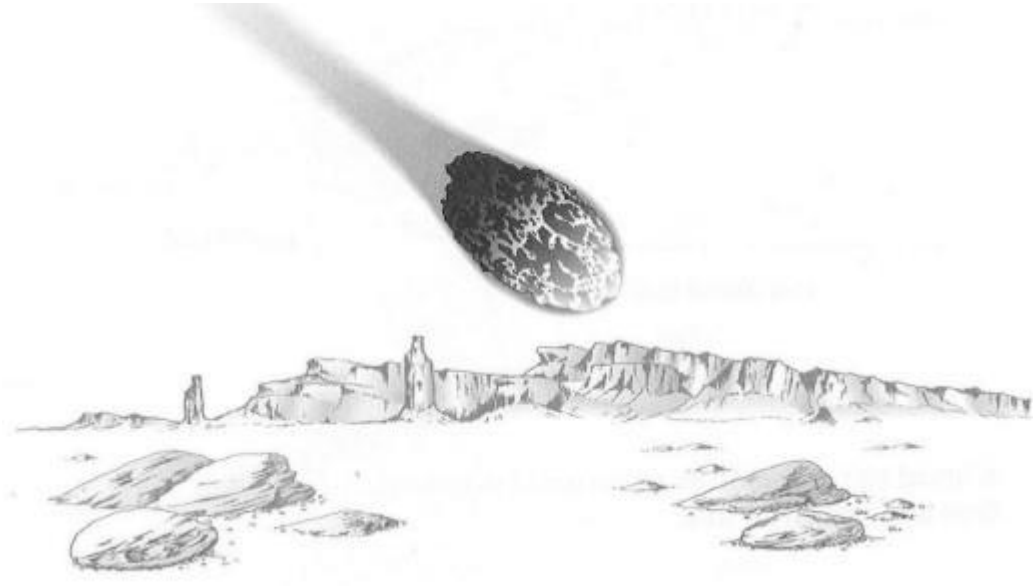
Q1. A meteorite landed on Earth. It contained a new element.
Scientists called the element jovium.



(a) The list below shows some properties of jovium.

Which **two** properties suggest that jovium could be a metal?
Tick **two** boxes.

It has a high melting point. ☐

It does **not** stick to a magnet. ☐

It is a blue solid. ☐

It is a good conductor of heat and electricity. ☐

It glows in the dark. ☐

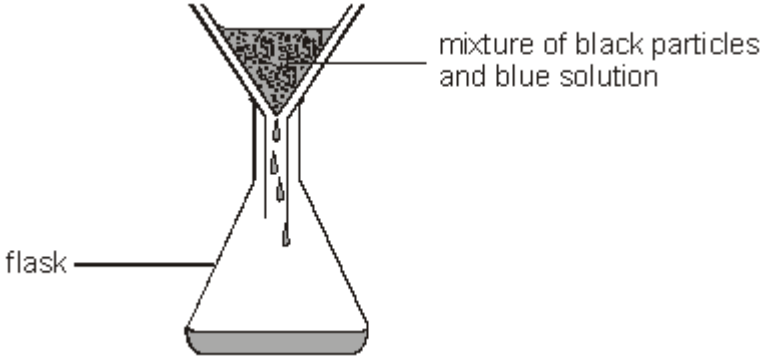
2 marks

(b) A scientist put a piece of the meteorite in water and stirred it.
This produced a blue solution with tiny, solid, black particles in it.

He separated the black particles from the blue solution using the apparatus below.

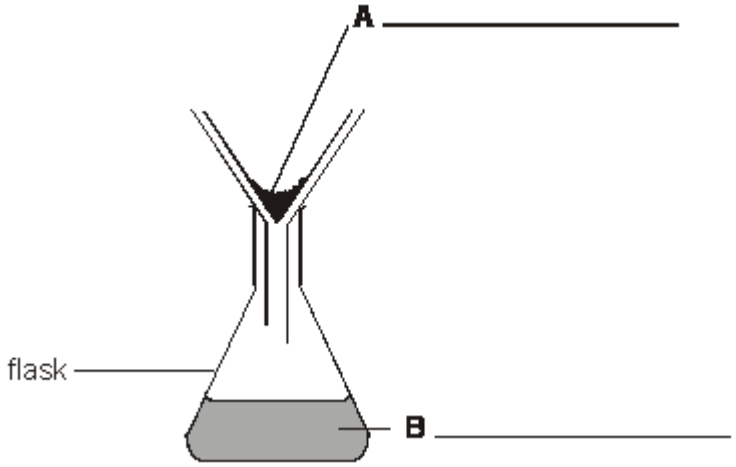
(i) Give the name of this method of separation.

.....



1 mark

(ii) The diagram below shows the results.
What do the labels A and B show? Write your answers on the lines.



2 marks

(c) The scientist poured the contents of the flask into a dish.
Two days later there were blue crystals in the dish, but **no** liquid.



at the start **two days later**

What happened to the liquid in the dish?

.....

1 mark
maximum 6 marks

Mark schemes

Q1.

(a) It has a high melting point. ✓

1 (L4)

It is a good conductor of heat and electricity. ✓

if more than two boxes are ticked,
deduct one mark for each incorrect tick
minimum mark zero

1 (L4)

(b) (i) filtration **or** filtering

accept 'filter'

1 (L3)

(ii) *A*: black particles **or** solid **or** residue

*accept 'particles' **or** 'solids'*
*do **not** accept 'blue particles'*

1 (L3)

B: blue solution **or** filtrate

accept 'solution'
accept 'liquid'
*do **not** accept 'black solution'*
*do **not** accept 'mixture'*

1 (L3)

(c) the water had evaporated

*accept 'the liquid had evaporated' **or** 'it had evaporated'*
'it had dried up' is insufficient
accept a description of evaporation eg 'it turned into gas'
*do **not** accept 'the solution evaporated'*

1 (L4)

[6]